Keycloak server as a GA4GH Passport broker

Leveraging GA4GH Passport for handling data permissions among different data repositories in the scope of the iPC project.

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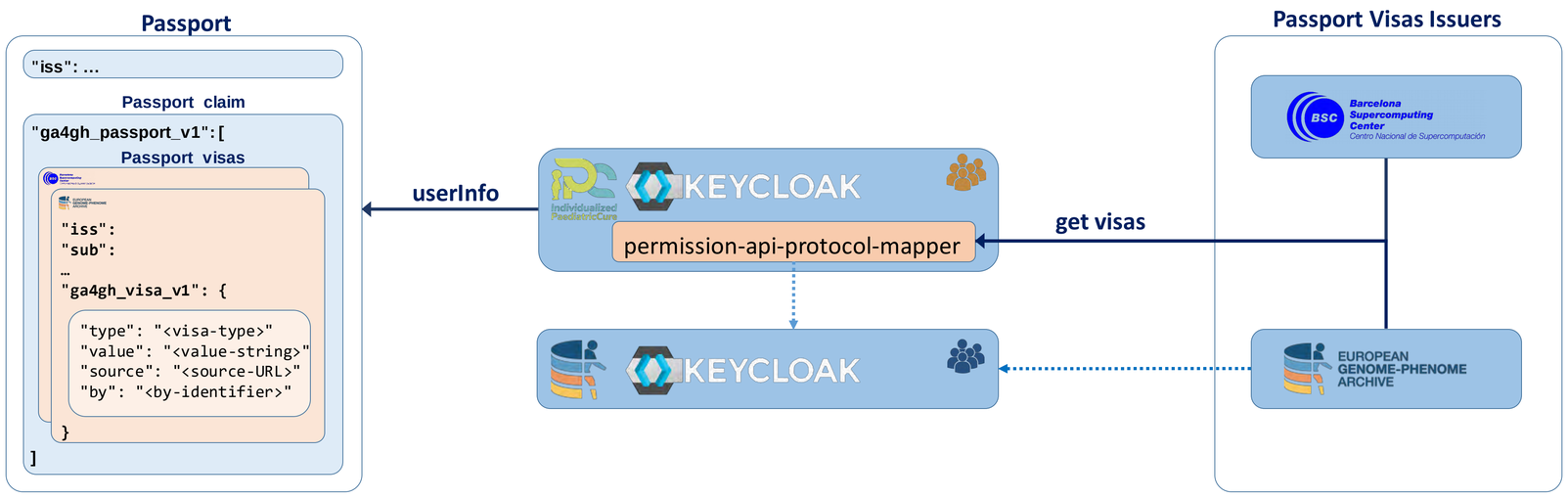
The Individualized Paediatric Cure (iPC) project is multi-institutional international effort that aims to improve paediatric cancer treatment using state-of-the-art computational methods and models. To provide better therapies for kids with cancer there is a need of accessing data from sparse, heterogeneous resources. These resources usually belong to different systems where multiple user credentials and specific data access policies are often required. To achieve interoperability across institutions, some platforms that host and disseminate cancer research data are adopting OpenID Connect (OIDC) protocol and The Global Alliance for Genomics and Health (GA4GH) Passport.

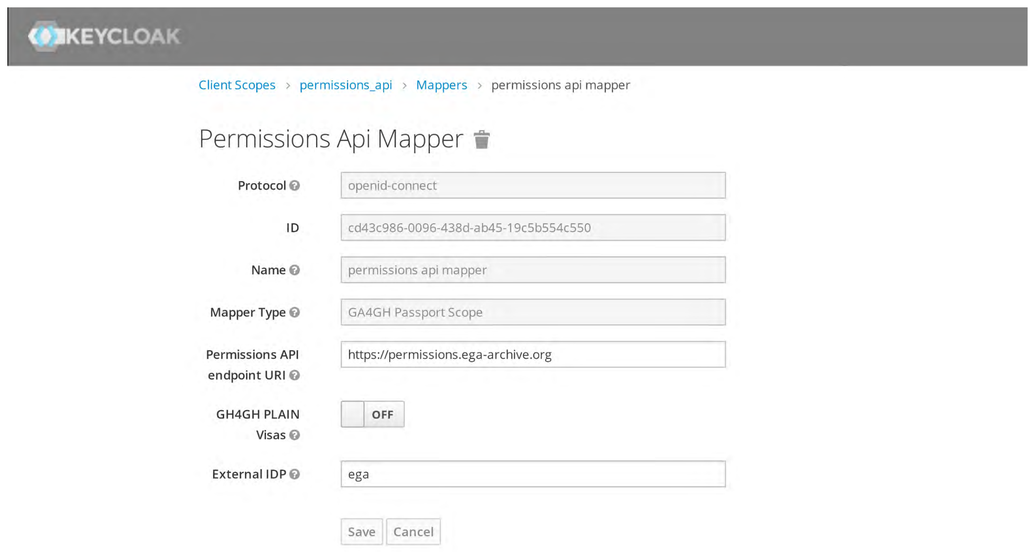
Here is shown the solution developed for achieving interoperability between the iPC platform and the European Genome-Phenome Archive (EGA) based on these standards. Specifically, a Keycloak OIDC server has been deployed and adapted to work as GA4GH Passport Broker.

Keycloak server as a GA4GH Passport broker



The principal role of the GA4GH Passport Broker is to issue GA4GH Passports – the OAuth2 JWT access tokens with "ga4gh_passport_v1" claim which embeds GA4GH Visas. These Visas are issued by the Visa Issuers – the services that provide signed embedded token with information about the user's affiliations, roles, accepted terms, grants, etc. GA4GH Passport specification doesn't specify the protocol between GA4GH Brokers and GA4GH Visa Issuers. European Genome Archive proposed REST-based Permission API protocol for this communication which is supported by the Resource Entitlement Management System (REMS) system in the ELIXIR AAI platform.

The **permission-api-protocol-mapper** is implemented as a Keycloak OpenID Connect Protocol mapper extension. It provides ¨ga4gh_passport_v1¨ scope support aggregating GA4GH Visas emitted by the Visas Issuers that support EGA REST Permission API.



https://gitlab.bsc.es/inb/ga4gh/ga4gh-passport/permission-api-protocol-mapper

```
{
  "sub": "a855e990-9be2-413d-9171-3d74a8360756",
  "aud": "ipc-rest",
  "iss": "https://inb.bsc.es/auth/realms/IPC",
  "ga4gh_passport_v1“: [
    {
      "format": "PLAIN",
      "iss": "",
      "sub": "1201754",
      "jti": "urn:uuid:7f9bde80-9c74-4f59-a6dd-37a91944581a",
      "ga4gh_visa_v1": {
        "type": "ControlledAccessGrants",
        "asserted": 1647255595,
        "value": "EGAD00000000028",
        "source": "EGAC00001000514",
        "by": "dac“
      }
    }
  ]
}
```

The extension also supports identity exchange as Visa Issuers most probably would be protected via their own Identity Providers. In order to provide identity exchange capability, Keycloak Realm must be configured to store OIDC offline token (¨offline_access¨ scope) obtained in the first external Identity Provider login and identity linking process. This token allows the mapper to refresh external Identity Provider's access token to pass with Permissions API REST request.

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Individualized PaediatricCure

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